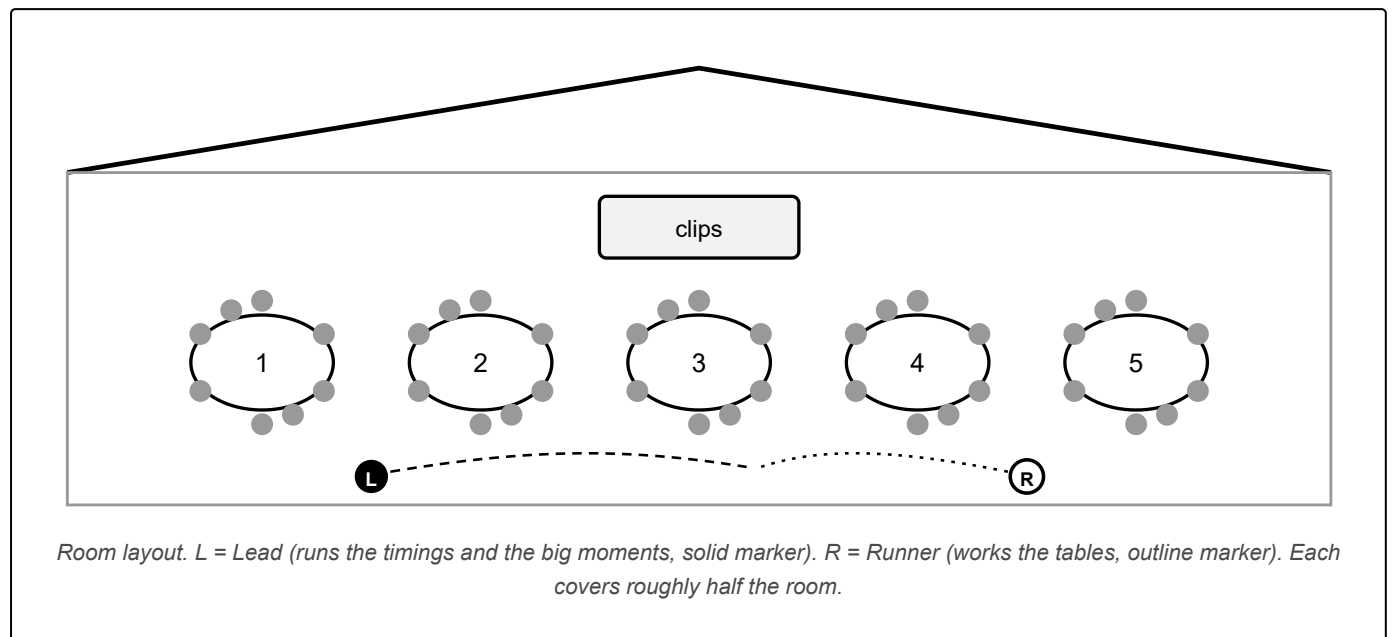


Who lives on a seaweed farm?

A one hour, hands-on workshop for adults. Build the food web of a seaweed and shellfish farm on the table in front of you, then take the farm away and watch what happens.

60	30-40	5	2
minutes	participants	tables of 8	facilitators



• THE ONE THING TO GET RIGHT

Everything builds towards minute 24, when you physically remove the farm. Protect that moment. If you are running late, cut the discussion, never the removal.



KIT LIST

- ☐ 5 × A1 laminated farm mats
- ☐ 5 × deck of 8 species cards (40 total)
- ☐ 1 spare deck (for a sixth table or losses)
- ☐ 12 × wipe-clean pens, plus cloths
- ☐ Flipchart pad and 4 chunky markers
- ☐ Laptop, charged, clips downloaded offline
- ☐ Printed stills of the 3 clips as backup
- ☐ Mussel shells, a metre of dropper rope
- ☐ Fresh kelp, collected that morning
- ☐ 2 × A5 crib sheets (page 9)

▲ WATCH OUT

Test the pens on the laminate before the day. The wrong marker is permanent and you will destroy £100 of mats in the first ten minutes. Buy wipe-clean, not dry-wipe-adjacent, and take a cloth per table.

THE STORY YOU ARE TELLING

BEAT	WHAT THE ROOM LEARNS
Build the web	A seaweed and shellfish farm is not empty water. It is crowded.
Remove the farm	A third of that life was only there because of the farm.
The sea potato	It is not all good news, and we say so first.
Costs and gives	This is a trade-off worth measuring, which is why PEBL films it.

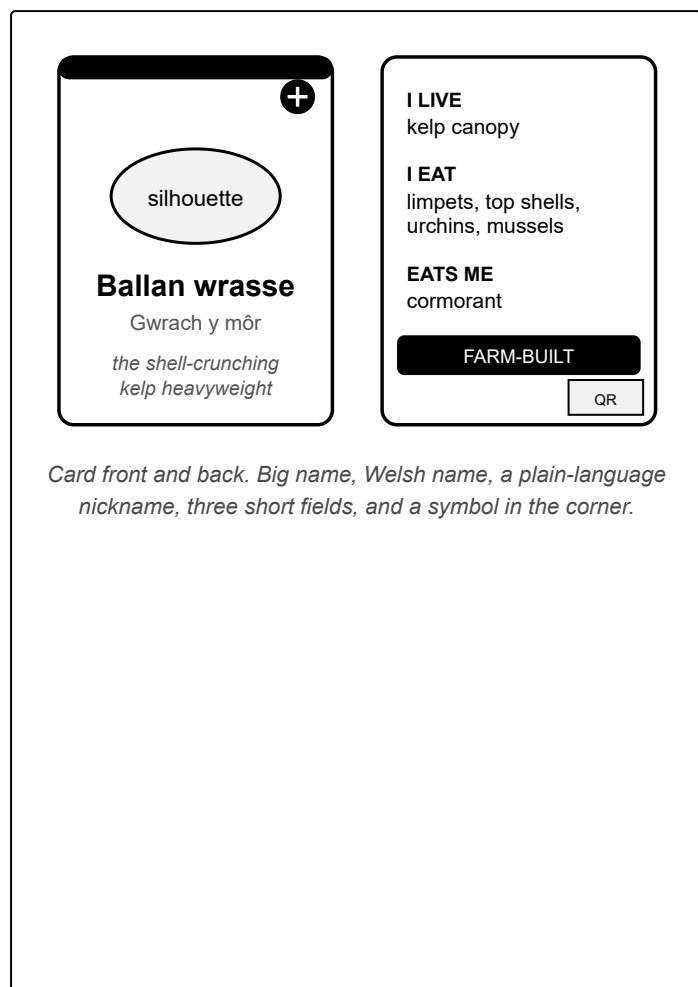
• THE NUMBERS, IF YOU ARE ASKED

Across the full catalogue we map 72 species and 238 feeding links on a farm. Take the farm away and it is 47 species and 133 links. The farm builds roughly a third of the species and just under half the feeding connections. Say "roughly a third" out loud, not the precise figures.

WHAT TO PRINT

ITEM	SPEC	COST
Farm mats ×5	A1, colour, encapsulated (gloss). Farm cross-section with the four food sources printed on it, no animals	~£100
Species cards ×80	A6, 350gsm, double sided, colour. Two identical sets of 40 so the activity survives being given away	~£90
Wipe pens, cloths	Fine tip, non-permanent	~£15
Flipchart and markers	Plus tape for the wall	~£25
Props and contingency	Shells, rope offcut, kelp on the day	~£25
Total		~£255

The mats and species cards above are the only colour items — they go to a proper print shop. This guide and the evidence cards are built for an ordinary black & white office printer or photocopier.



CARD RULES

- **Name at 14pt minimum.** A marquee is dim and half your room is over 60.
- **Welsh name under the English.** You are in Tyddewi. This is cheap and it matters.
- **A nickname,** so nobody has to admit they have never heard of a dragonet.
- **Never colour alone.** Each status gets a symbol as well as a colour.
- **Max three fields on the back.** If it needs a fourth, cut it.
- **QR on every card,** to FishSpotter. The card is the takeaway, so there is no leaflet.

THE FOUR SYMBOLS

• **FARM-BUILT** Gone without the farm

● **BOOSTED** Fewer without it

○ **HERE ANYWAY** Barely changes

+ **BETTER WITHOUT** One card only

▲ WATCH OUT

Do not laminate the species cards. People take them home and a laminated card feels like something to hand back. 350gsm card is warmer, cheaper, and posts through a letterbox.

People are still arriving. Do not wait for them. Start with the clips playing and talk over the top.

Play three short clips filmed on a working Welsh farm: a pollack, a spider crab on a rope, a cuttlefish. Ask the room to call out what they can see. Take every answer, including the wrong ones.

“ SAY ROUGHLY THIS

"Everything you have just watched was filmed on a seaweed and shellfish farm, about an hour up the coast. Nobody put those animals there. They turned up."

▲ GET IN FRONT OF THE SALMON QUESTION

Somebody in this room thinks a marine farm means caged salmon, sea lice and chemicals. If you let that sit unspoken you will spend the hour under suspicion. Say it yourself, now, before anyone asks:

"Quick thing, because people always ask. Nothing here is fed, caged or medicated. Seaweed and mussels take food out of the water rather than having it poured in. It is the opposite of a salmon farm. If you want to get into that, grab me at the end."

Runner deals one deck per table, face up, and one mat per table. Everyone takes a card. Late arrivals slot into any table with a spare card.

“ SAY ROUGHLY THIS

"Take a card. You are now in charge of that species. Have a read. You need to know three things: where you live, what you eat, and what eats you."

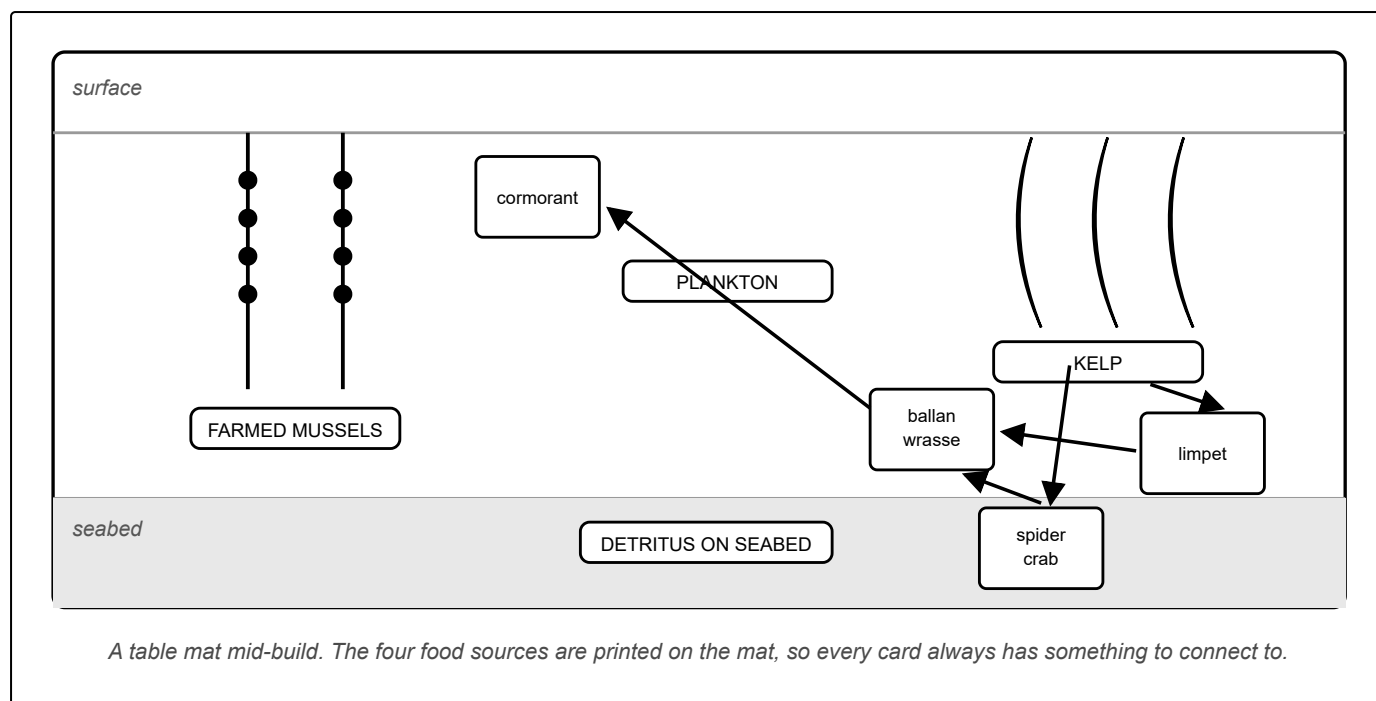
• WORDING MATTERS

Say "you are in charge of the cuttlefish", not "you are a cuttlefish". Adults will happily be an expert on an animal. They will not happily pretend to be one, and the difference costs you nothing.

• IF SOMEONE WILL NOT TAKE A CARD

Fine. Give them the job of counting arrows for the table. They are now the most useful person there, and they will usually pick up a card within ten minutes anyway.

Tables place their cards on the mat by where the species lives, then draw an arrow from each meal to whoever eats it. Arrows point the way the food travels.



• WHY THE DECKS ARE PRE-SORTED

Each table has been given a set of eight species that genuinely eat each other. Do not shuffle the decks together or let tables swap cards. If you do, tables end up drawing arrows to species that are sitting on a different table, and the activity stalls.

PROMPTS THAT UNSTICK A TABLE

- "Start with the food sources printed on the mat. Who eats straight off those?"
- "Right, who at this table would eat *you*?"
- "Follow it up. Who eats the thing that eats you?"

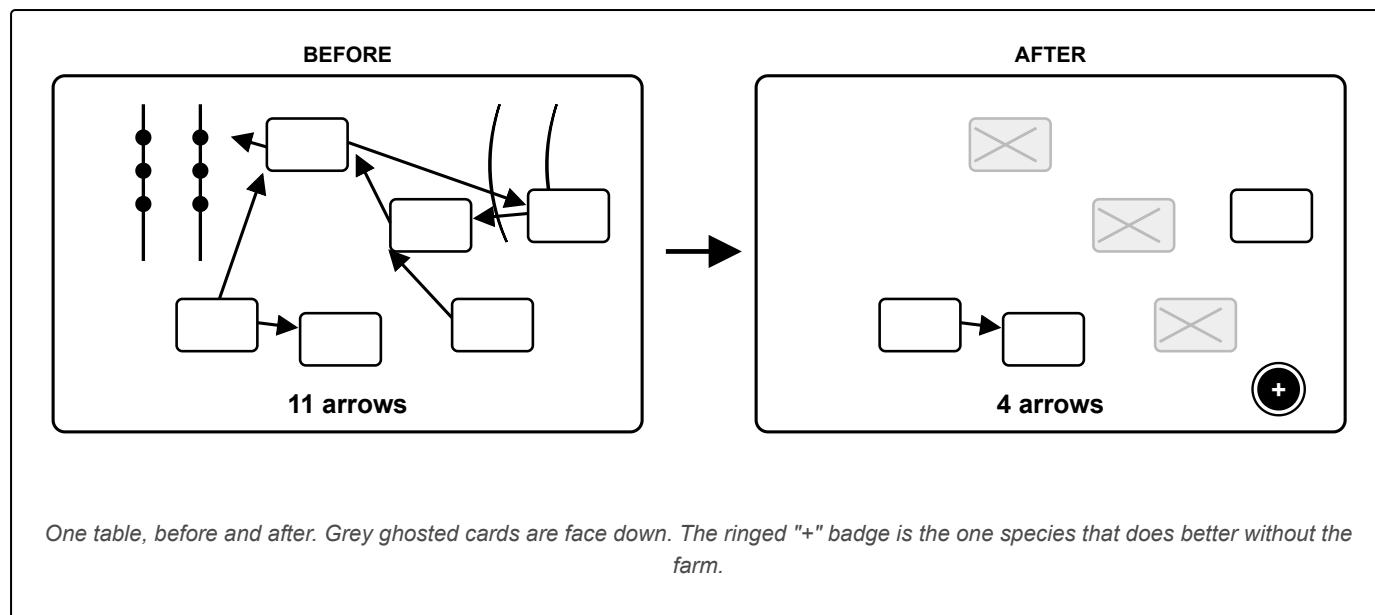
PACE

- **Fast table:** "Count your arrows and write the number in the corner. Now find the longest chain you can."
- **Stalled table:** sit down with them and place two cards yourself. Do not narrate, just start.

▲ EVERYONE MUST WRITE THEIR ARROW COUNT ON THE MAT BEFORE MINUTE 24.

You need that number for the reveal. Runner walks the room at minute 22 and checks every table has one.

This is the whole workshop. Ninety seconds of it. Direct it properly: one voice, everyone quiet, no side conversations.



RUN IT IN THIS ORDER

1. **Stop the room.** "Pens down, everyone look up." Wait for actual quiet.
2. **Both facilitators move in.** Lead counts down from three. On "one", both of you physically lift the kelp and the mussel rope props off the tables and put them on the floor, out of sight.
3. **Beat of silence.** Do not fill it.
4. "If your card has the filled circle, the farm was the only reason you were here. **Turn your card face down.**"
5. "Now cross out every arrow that touches a face-down card."
6. **Go round the tables out loud.** "Table one, what were you and what are you now?" Let them say the numbers themselves. Table 1 will report something like eleven down to three.
7. **Then the plus card.** "Is anyone better off?" Table 5 has the sea potato. Get that person to hold it up.

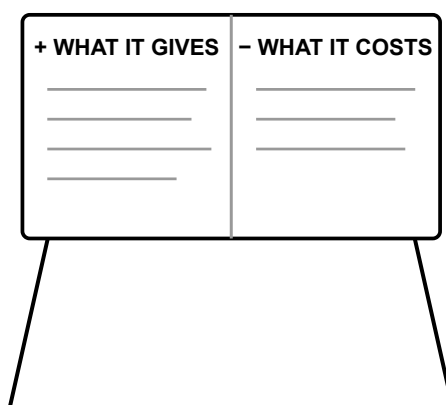
• WHY TABLE 5 MATTERS

The tables are deliberately uneven. Table 1 loses five of its eight and is nearly wiped out. Table 5 loses one, and gains a species that does better once the farm stops raining material onto the sand. That spread is the honest picture, and the room notices it without you having to say it.

▲ DO NOT RUSH TO THE MORAL.

The temptation is to explain what just happened. Do not. Ask "so what did that just show you?" and let a member of the public say it out loud instead. It lands ten times harder in their words.

Two columns on the flipchart. The room fills them in. Your job is to write, not to argue.



Keep it on one sheet so the whole room can see both columns at once.

WHAT USUALLY COMES UP

Gives. Shelter and nursery space for young fish. Food for crab and lobster. Somewhere for things to settle. Calmer water inside the lines. Food produced without feed, fresh water or fertiliser.

Costs. Sea space that other people used. Gear and navigation conflict. Enriched mud under the lines that suits fewer clean-sand species. The habitat disappears when the crop is harvested. Ropes and floats need maintaining or they become litter.

• HAND THE PEN TO THE SCEPTIC

If there is a fisherman in the room, and at this festival there will be, invite them to fill in the costs column themselves. It is the single highest value thing you can do all hour. It turns the person most likely to dismiss you into the person who helped write the honest version, and everyone else sees you do it.

“ SAY ROUGHLY THIS TO CLOSE THE STEP

"We are not here to tell you a farm is all upside. We are here because nobody has actually measured this properly in Wales, and that is the job we are doing. Both columns are real."

▲ DO NOT LET THIS BECOME A PLANNING MEETING.

If two people start relitigating a specific local licence application, take it offline: "That is a proper conversation and I want to have it, can we do it at the end so we get through the rest?" Then genuinely do it at the end.

Once both columns are full, hand each table its **evidence card** from the separate pack. One person per table reads the headline out loud. Three minutes, and it changes the register of the room completely, because the session stops being your opinion and starts being a measurement.

TABLE	THEIR CARD	THE FINDING
2 ropes	The reef that came back after 150 years	30% more reef in 7 years, Lyme Bay
3 shoal	More fish, and young ones settling in	1.6× more fish, larvae of 7 species
4 seabed	Crab and lobster do well out of it	Commercial crustaceans gain
1 canopy	A patch of easier water for shellfish	23% easier shell-building
5 sand	A farm is not a wild kelp forest	Fewer species than natural habitat

MAKE THEM GUESS FIRST

Before table 2 reads their number, stop them and ask the whole room:

“ SAY ROUGHLY THIS

"They watched the seabed under a mussel farm in Lyme Bay for eight years, against control sites nearby. Hands up who thinks the seabed got better. Right, how much more reef after seven years? Shout me a percentage."

Take three guesses, then let table 2 read out **30%**. A number you have guessed at is a number you remember. This is the cheapest engagement trick in the guide and it works on adults every time.

• READ TABLE 5'S CARD LAST, AND DO NOT SOFTEN IT

There is the study that found kelp farms hold *fewer* species than natural habitat nearby. Ending the evidence round on the finding that works against you is the strongest thing you will do all hour. Follow it with: "which is why we think farms belong on bare ground, not on top of something that is already good."

▲ TWO CARDS ARE YOURS, NOT THE TABLES'

The carbon card and the PEBL card stay with the facilitator. Use the carbon card the moment anyone mentions carbon, and the PEBL card to close the session.

Print this page at A5 and carry one each. These four come up nearly every time.

1. "ISN'T THIS THE SAME AS A SALMON FARM?"

No, and it is the opposite in one specific way. Salmon are fed, so a salmon farm puts food, nutrients and sometimes medicines into the water. Seaweed and mussels are not fed anything. They take nutrients out of the water to grow. Different industry, different footprint, and it is fair to judge them separately.

2. "DOESN'T SEAWEED LOCK UP LOADS OF CARBON?"

It grows fast and it absorbs carbon dioxide while it grows. But if we harvest it and eat it, or put it on a field, or in a bottle of shampoo, that carbon comes back out fairly quickly. Only the fraction that sinks to deep water and stays there counts as storage, and nobody has measured that well yet for UK farms. The strong climate case here is habitat and shelter, not storage.

• SAY THIS EVEN WHEN NOBODY ASKS

Somebody in the room will confidently tell their neighbour that seaweed farms are a big carbon solution. Correcting it gently, in public and early, is the single biggest credibility win available to you. People trust an organisation that argues against its own hype.

3. "THAT'S MY FISHING GROUND YOU ARE COVERING UP."

Straight answer: yes, a farm takes sea space, and that is a genuine cost to somebody. What we can say is that the structure feeds and shelters things that crab and lobster eat, and young fish use it as cover. Whether that is worth the space you lose depends on where it goes and how big it is, which is exactly why we put cameras on them instead of guessing.

▲ DO NOT OVERSELL TO A FISHERMAN.

They will know more than you about that stretch of coast. Ask where they work and what they see on their gear. Treat it as data collection, because it is.

4. "WILL THIS HELP WITH THE SEA WARMING UP?"

Not the temperature itself. It helps things cope. The lines take the energy out of waves so there is calmer water inside. Seaweed photosynthesising raises the pH of the water right around it, which matters because the sea is slowly acidifying and that makes it harder for young mussels, crabs and other shellfish to build their shells. A farm is a small patch of easier conditions.

5. "WHO PAYS YOU?"

PEBL is a community interest company, not a private business with shareholders. Say who actually funded the work, name them, and move on. Evasiveness here costs you the room.

Each table gets eight cards that genuinely eat one another. Keep the decks separate. ● farm-built, ① boosted, ○ here anyway, + better without.

Table 1 · The kelp canopy ● loses 5 of 8

Seaweed feeds the grazers, the grazers feed the wrasse, the wrasse feeds the cormorant.

- ● Common limpet
- ● Edible sea urchin
- ● Painted top shell
- ● Ballan wrasse
- ● Two-spotted goby
- ① Spider crab
- ○ Thick-lipped mullet
- ○ Great cormorant

Table 2 · The mussel ropes loses 3 of 8

Everything here is queueing for the same meal.

- ● Dog whelk
- ● Velvet swimming crab
- ● Spiny starfish
- ① Common starfish
- ① Common brittlestar
- ① Common eider
- ○ Shore crab
- ○ Edible crab

Table 3 · The rope shoal loses 3 of 8

Plankton to sprat to pollack to a diving bird, in four steps.

- ● Fifteen-spined stickleback
- ● Poor cod
- ● Bib
- ① Pollack
- ① Atlantic cod
- ○ Sprat
- ○ Sand smelt
- ○ European shag

Table 4 · The seabed crew ● loses 5 of 8

What falls from the farm feeds the bottom, and something big lives in the anchor blocks.

- ● Butterfish
- ● Shanny
- ● Rock goby
- ● Curled octopus
- ● Conger eel
- ○ Hermit crab
- ○ Lesser-spotted catshark
- ○ Grey seal

Table 5 · The open sand ★ loses 1, and one does better

The table that barely changes. Give this deck to the group that looks most sceptical.

- + Sea potato
- ● Long-spined sea scorpion
- ○ Purple heart urchin
- ○ Dragonet
- ○ Sand goby
- ○ Plaice
- ○ Flounder
- ○ Harbour seal

• DEAL TABLE 5 DELIBERATELY

It is the quietest table during the reveal and the most important one during the discussion. Put your most talkative or most sceptical group there, because they are the ones who will say "well, hang on, we barely changed" out loud, which is exactly the honesty the session needs.

“ SAY ROUGHLY THIS

"Keep your card. The code on the back opens the footage our cameras are collecting on real farms. If you want to know what actually lives on one, you can sit at home and help us identify it. That is not a gimmick, we genuinely use the identifications."

Leave five minutes of slack at the end. You will use it. The best conversations of the hour happen after you have officially finished, usually with the person who said the least.

IF IT GOES WRONG

PROBLEM	WHAT TO DO
Only 12 people come	Run tables 1, 4 and 5 only. That keeps the wipeout table, the seabed table and the sceptic table. The reveal still works.
50 people come	Deal the spare deck as a sixth table and pair people on cards. Do not add a facilitator job you do not have.
No power or wifi	Clips are already downloaded on the laptop. If the laptop dies, use the printed stills and describe the footage. The activity does not depend on the screen.
Running 10 minutes late	Cut step 5 to three minutes and take questions in the slack. Never cut step 4.
A table finishes very early	"Find the longest food chain on your mat." Then, "which of you would still be here without the farm? Predict it before we do the reveal."
Someone dominates	"Let me get a couple of other views and come back to you." Then actually come back to them.
Nobody talks at all	Go table by table rather than asking the open room. Ask the table, not an individual.

AFTERWARDS, FIVE MINUTES WHILE YOU PACK UP

- ☐ Photograph one completed mat before wiping it
- ☐ Note the two questions you could not answer well
- ☐ Rough headcount and how many cards went home
- ☐ Wipe mats before the pens set

• THE MEASURE OF SUCCESS

Not how many people came. Whether the most sceptical person in the room left still talking to you. If a local fisherman filled in the costs column and stayed behind to argue about siting, that was a good workshop.

Plant Ecology Beyond Land (PEBL) CIC · company number 12076622 · www.pebl-cic.co.uk · hello@pebl-cic.co.uk
 Species, diets and farm impact generated from the FishSpotter catalogue and the PEBL farm food web. A teaching schematic, not a quantified survey.